Sandoval County Groundwater Data | Research Brief

NMBGMR groundwater-level dataset • Guidance for project plots and interpretation

Dataset overview

The attached CSV contains 456 records for 87 distinct well identifiers, with measurement dates from 1997 to 2019. Useful fields include PointID, Easting, Northing, WellDepth, HoleDepth, DateMeasured, and DepthToWaterBGS (depth to water below ground surface). Multiple records can refer to the same well.

Research questions best supported

Q2 — Well records	**Strongest fit (partial): compare recorded well depths and depth to groundwater; aquifer formation is not identified in this CSV.**
Q8 — Monitoring coverage	Strong: map well coordinates and summarize number of wells, measurement dates, and spatial/temporal gaps.
Q1 — Groundwater trends	Moderate: plot repeated depth-to-water readings for wells with enough observations; one-off readings cannot establish trends.
Q9 — Dashboard indicators	Moderate: build well-location maps, depth distributions, observation counts, and qualified trend summaries.

Relevance to Cuba's municipal water sources

NOT DIRECTLY ESTABLISHED. This countywide dataset does not identify Cuba's municipal supply wells or the aquifer formations supplying them. Previous geographic review indicated that the closest recorded well is roughly 24 miles from Cuba. Therefore, do not use these measurements to claim that Cuba's municipal aquifer is declining or nearing depletion. They remain useful for countywide comparisons and identifying where Cuba-specific monitoring is missing from this dataset.

Suggested plots for the project report

- Map of unique well locations (Easting/Northing; confirm UTM datum and zone before mapping).
- Scatterplot of WellDepth versus DepthToWaterBGS, excluding missing values and clearly distinguishing different wells.
- Time-series plots of DepthToWaterBGS by DateMeasured for wells with repeated observations.
- Bar chart of measurement counts per well and a timeline of monitoring coverage.

Interpretation cautions

Increasing DepthToWaterBGS means the water level is getting deeper (falling), not rising. Do not treat all 456 rows as independent wells; check duplicate readings, missing values, measurement methods, and data-quality flags. Differences between wells may reflect distinct aquifers and elevations. For direct Cuba conclusions, obtain municipal well IDs, aquifer identifications, pumping records, and groundwater-level histories.

Source: https://catalog.newmexicowaterdata.org/dataset/groundwater-levels/resource/de88e71a-3f5b-4bc9-8fb0-28b81c198952

Source: NMBGMR groundwater-level CSV supplied for Sandoval County